Stockholm, Sweden
mikael@spacecorpstechnology.com
github.com/rorychatt
linkedin.com/in/mikael-rinne
rorychatt.github.io/rorychatt

# Mikael Rinne

**Founding Engineer · Rust, C#, TypeScript**

## PROFILE

Founding engineer who builds developer tools and real-time systems in **Rust, C# and TypeScript**. Top contributor to **Ivy Framework** (437 GitHub stars) and lead engineer of **Ivy Tendril**, a development environment for teams where AI agents write most of the code. Founder of **SpaceCorps**, which ships a space MMO on its own Rust engine and maintains 20+ open-source Rust CLIs for AI agents. Started in computational physics and machine learning, with a first-author IEEE paper written with CERN collaborators.

## EXPERIENCE

### **Ivy Interactive** · 1st Founding Engineer

2025 – present · Stockholm

*A .NET AI startup: a pure-C# full-stack framework for internal tools with LLM code generation, and Tendril, an agentic development environment.*

- Top contributor to **Ivy Framework** (C#/.NET, 437 stars): **3,586 commits and 1,148 PRs**, 38% of the repository's history.
- Lead engineer of **Ivy Tendril** since its first commit in April 2026 (203 stars; macOS, Windows, Linux): **2,128 commits and 652 PRs**, 48% of the repository. Agents run in parallel in isolated git worktrees and changes are reviewed and merged in-app. Now leading the V2 rewrite as a TypeScript and Rust monorepo.
- Wrote much of the Rust layer: Rusty-Framework (pure-Rust port of Ivy), Ivy.NativeJsonDiff (RFC 6902 diffing over Rust FFI for reactive UI updates), Ivy.Docs.Compiler, Tendril-Service (headless agent orchestration service and CLI), and co-built Rustino (native webview windows, a Photino.NET replacement).
- Own product engineering, AI modelling and planning, and lead a team of developers. Across the company's GitHub organisation: **2,700+ PRs authored and 600+ reviewed**.
- Speaker at the Agentics Foundation "Harness Engineering" meetup (IBM Innovation Studio) as maintainer of Ivy Tendril; author of the talk "Three things that nearly killed our software factory".

### **SpaceCorps Technology OÜ** · Founder & CEO / CTO

2023 – present · Remote, Europe

- Founded the company around an open-source 3D MMO; recruited and lead an international team of six (designers and full-stack developers). Built the original WebGL prototype, SpaceCorps 2 (TypeScript, three.js), myself.
- Built **Space3d**, an in-house game engine in Rust on wgpu/WebGPU targeting native, web and mobile, and **SpaceCorps 2027** on top of it: a space MMO with public builds for Windows, macOS and Linux (latest build v0.5.1).
- Published **20+ MIT-licensed Rust CLIs** that give AI agents safe access to real services (Gmail that can only draft, Cloudflare, Firecrawl, Exa, Sliplane, Namecheap and more): structured output, stable exit codes, OS-keychain credentials and prompt-injection guards. Co-built with Niels Bosma.
- Open-sourced open-apple-models and open-android-models (Swift, Kotlin; pre-release): tool calling for on-device LLMs (Apple Foundation Models, Gemini Nano) to drive game NPCs.

### **Scania** · Solutions Architect

2024 – 2025 · Stockholm

- Developed new modules for Scania's internal core platform, with a focus on stability, user experience and CI/CD workflows.
- Contributed to the move to a newer front-end stack (Angular 19).

### **Nixor EE AS** · Full-stack Software Engineer

2023 – 2024 · Tallinn

- Built and shipped SaaS and POS improvements for Coop, Stockmann Group and Selver, using custom Odoo modules and payment integrations.
- Found and fixed critical transaction bugs in production retail systems across Estonia.

### **University of Tartu, Institute of Technology** · Software Engineer & Researcher

2022 – 2023 · Tartu

- Built scientific web software for molecular dynamics and quantum emission calculations with CERN collaborators (GETELEC, 174 commits), bridging theoretical and experimental physics.
- First author of "An Online Tool for Thermal-Field Emission Calculations from Metal and Semiconducting Emitters", IEEE IVNC 2023.

**Private tutor, natural sciences** 2019 – 2024 · Tallinn, remote

- Taught natural sciences one-on-one, in groups and remotely, helping students improve their academic results. Mentoring is still central to how I run a team.

## OPEN SOURCE

**Ivy Framework** · C#/.NET, 437 stars
github.com/Ivy-Interactive/Ivy-Framework

**Ivy Tendril** · agentic dev environment, 203 stars
github.com/Ivy-Interactive/Ivy-Tendril

**SpaceCorps CLI fleet** · 20+ Rust CLIs for agents, MIT
github.com/SpaceCorps

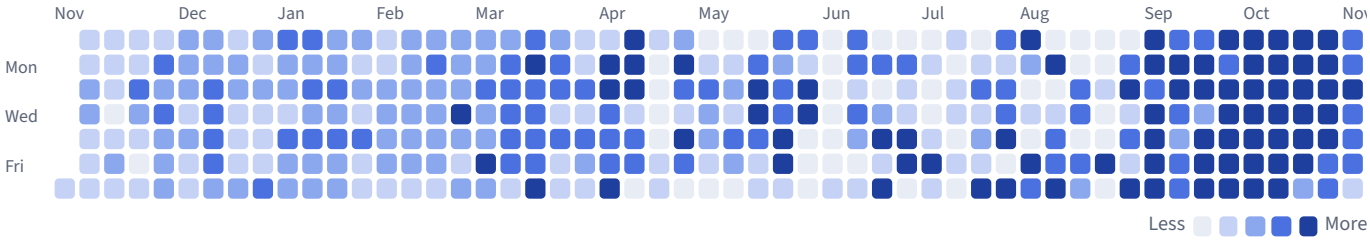
**open-apply** · Rust CLI for AI-assisted job search and application tracking, MIT
github.com/SpaceCorps/open-apply

**open-apple-models** · on-device LLM tool calling, Swift, MIT
github.com/SpaceCorps/open-apple-models

**Rusty-Framework** · full-stack web framework in pure Rust
github.com/Ivy-Interactive/Rusty-Framework

## GITHUB ACTIVITY, LAST 365 DAYS

**29,603** contributions in the last 365 days · **317** active days · longest streak **140** days




## EDUCATION & HONORS

**SALT School of Applied Technology** 2024
Full-stack C# developer programme: TDD, mob programming, ASP.NET, Blazor, EF Core, React.

**University of Tartu, Chemistry** 2019 – 2024
Studied chemistry with a heavy computational and data science focus: ML models for molecular dynamics (Gaussian approximations, Monte Carlo, neural networks, gradient boosting) and simulation in COMSOL, Modelica and Fortran.

**Bronze medal, 51st International Chemistry Olympiad** 2019

**1st place, Estonian National Chemistry Olympiad** 2018

**Google: Foundations of Data Science, Python** 2024

## SKILLS

| | |
|---|---|
| **Languages** | Rust, C#, TypeScript, Go, Python, SQL; also Swift, Kotlin, C/C++, Java, Fortran, WGSL |
| **Backend & cloud** | .NET 10, ASP.NET Core, EF Core, SignalR, axum, Tokio, PostgreSQL, Docker, Kubernetes, AWS, Azure, GCP, GitHub Actions |
| **Frontend & graphics** | React, Vite, Tailwind, TanStack, three.js/WebGL, wgpu/WebGPU |
| **AI engineering** | Agent orchestration, tool calling, MCP, Anthropic and OpenAI APIs, RAG and vector search, on-device models, evals |
| **Also used** | Angular, Node.js, Dapper, Odoo, SQL Server, MySQL, xUnit, Vitest, Jest, Jira, Confluence, COMSOL, Modelica, MATLAB, Assembler |
| **Practice** | TDD, SOLID, agile, CI/CD, code review, mob programming, technical leadership, hiring and mentoring |
| **Human languages** | English (native), Estonian (fluent), Russian (fluent) |